MCP Basics

Model Context Protocol explained simply

A 10-slide introduction

Agenda

- What MCP is
- Why it exists
- Main components
- How an interaction works
- Benefits and safety

What is MCP?

- MCP stands for Model Context Protocol

- It is an open standard for connecting AI applications to outside systems

- Those systems can provide data, tools, and reusable workflows

- It is often compared with a common connector such as USB-C

Why MCP exists

- AI applications often need access to files, calendars, databases, and services
- Custom integrations create duplicated work
- A shared protocol gives applications and external systems a consistent way to communicate
- Developers can reuse integrations across compatible applications

Main components

Host

The AI application

Client

The connection manager

Server

The outside helper

Tools, resources, and prompts

- Tools: functions the AI can call to perform actions
- Resources: data that can be read and added as context
- Prompts: reusable interaction templates
- Servers describe which of these capabilities they support

Typical interaction

- The AI application connects to an MCP server
- The client and server negotiate their supported capabilities
- The application discovers available tools or resources
- The AI selects a tool and supplies structured arguments
- The server returns a structured result

Benefits

- Less repeated integration code
- A growing ecosystem of compatible clients and servers
- A consistent method for discovering capabilities
- Support for local and remote servers
- Clearer boundaries between an AI application and outside systems

Security and control

- External actions and data access require careful permission design

- Remote connections can use standard authorization mechanisms

- Tokens should be issued for and validated by the intended server

- Applications should show users what data and actions are being requested

Summary

- MCP is a shared protocol for connecting AI applications to external systems
- Hosts connect through clients to servers
- Servers can provide tools, resources, and prompts
- The protocol standardizes communication, discovery, and structured results
- Good implementations still need permission, trust, and security controls